1. Binomial Distribution

- A binomial distribution is a **discrete distribution** that is modelled using two parameters
  - The number of **finite trials** n
  - The **constant** probability of success p
- The conditions for a model to follow a Binomial distribution:
  - **Finite** number of trials, denoted by n
  - Only **2 possible outcomes**, success and failure
  - Probability of success, p, is **constant** in each trial
  - Outcomes of each trial are **independent** of each other

1.1. Finding probability using the distribution

- We say a continuous random variable, X, follows a binomial distribution $B(n,p)$. This is written as

$$X \sim B(n,p)$$

The probability of the random variable X having x success in n trials can be found using:

$$P(X=x) = \binom{n}{x} \times p^x \times (1-p)^{n-x} \quad \text{[Given in MF19]}$$

Here:

- x is a positive integer.
  - x = 0, 1, 2, $\cdots$, n
- $(1-p)$ denotes the probability of a failure
  - It is also denoted by q

Mean and Variance

- To calculate the **expected value**, or the mean, of a random variable following a binomial distribution, we can use the formula

$$E(X) = \mu = np$$

- To calculate the **variance** of a random variable following a binomial distribution, we can use the formula

$$Var(X) = \sigma^2 = np(1-p)$$

1.2. Examples

{S11-P63} Question 6:

The probability that Sue completes a Sudoku puzzle correctly is 0.75. Sue attempts 14 Sudoku puzzles every month. XX denotes the number that she completes successfully. Find the value of X that has the highest probability. You may assume this value is one of the two values closest to the mean of X.

Solution:

Calculate the mean of X

$$E(X) = 14 \times 0.75 = 10.5$$

Successful puzzles completed have to be a whole number so they can either be 10 or 11.

$$P(10) = 14C_{10} \times 0.75^{10} \times 0.25^4 = 0.220$$

$$P(11) = 14C_{11} \times 0.75^{11} \times 0.25^3 = 0.240$$

Probability with 11 is higher, therefore, X = 11

{W11-P62} Question 6:

In Luttley College, 60% of students are boys. Students can choose exactly one of Games, Drama or Music on Friday afternoons. 75% of the boys choose Games, 10% choose Drama, and remainder choose Music. Of the girls, 30% choose Games, 55% choose Drama and the remainder choose Music. 5 drama students are chosen. Find the probability that at least 1 of them is a boy.

Solution:

First, we calculate the probability of selecting a boy who is a drama student, a conditional probability:

$$P(S) = \frac{P(\text{Boy}|\text{Drama})}{P(\text{Drama})}$$

$$P(S) = \frac{P(\text{Boy}) \times P(\text{Drama})}{(P(\text{Boy}) \times P(\text{Drama})) + (P(\text{Girl}) \times P(\text{Drama}))}$$

$$P(S) = \frac{0.6 \times 0.1}{(0.6 \times 0.1) + (0.4 \times 0.55)} = \frac{3}{14}$$

We can calculate the probability there is at least 1 boy present from 5 drama students using a binomial distribution with 5 trials and P of success = $\frac{3}{14}$

Find the probability of 0 and subtract the answer from 1:

$$P(X \geq 1) = 1 - P(X=0)$$

$$P(X \geq 1) = 1 - 5C_0 \times \left(\frac{3}{14}\right)^0 \times \left(\frac{11}{14}\right)^5$$

$$P(X \geq 1) = 0.701$$

1.3. Logarithmic inequalities

- Logarithms are a type of function that help us solve exponential equations

Given an equation

$$a^x = b$$

We can solve for x by using the logarithmic function

$$log_a(b) = x$$

- Most calculators have an inbuilt function which can solve logarithmic equations for you.
- If a logarithm is taken with a base, a, less than 1 then the **inequality is flipped**.

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- To solve inequalities in binomial distribution questions, its important to know that $\binom{n}{0} = \binom{n}{n} = 1$.

Example

A factory makes water pistols, 8% of which do not work properly.

In a random sample of n water pistols, the probability that **at least one** does not work properly is greater than 0.9. Find the smallest value of n.

Answer

We can model the distribution as such

$$X \sim B\left(n, 0.08\right)$$

From the question

$$P(X \geq 1) > 0.9$$

We can re-write this as

$$1 - P(X = 0) > 0.9 \implies P(X = 0) < 0.1$$

Using the distribution $X \sim B(n, 0.08)$

$$\binom{n}{0} \times (0.08)^0 \times (0.92)^n < 0.1$$

$$(0.92)^n < 0.1$$

As $0.92 < 1$, the inequality is flipped when we apply the logarithmic function

$$n > log_{0.92}(0.1)$$

$$n > 27.6$$

So the smallest value of n is 28.

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